1. Setup and Data Cleaning

1.1 Load libraries

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.1     v stringr   1.5.2
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(readxl)
library(cowplot)
```

```
##
## Attaching package: 'cowplot'
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

```r
library(fixest)
library(modelsummary)
```

1.2 Data cleaning

```r
DATA_PATH = "../data/01-raw/MINC3.xlsx"
mincome_data <- read_xlsx(DATA_PATH)

DATA_PATH_2 = "../data/01-raw/MINC4.xlsx"
payments_data <- read_xlsx(DATA_PATH_2, skip = 1, col_names = TRUE)
```

```
## New names:
## * `DI2` -> `DI2...6`
## * `DI2` -> `DI2...7`
```

```r
minorities <- c(16,19,7,8,20,21,24,26,32,33)

minc_clean <- mincome_data %>%
  mutate(
    HHHRWRK = MHHOURS + FHHOURS,
    RACIALHH = if_else( MHETHNIC %in% minorities | FHETHNIC %in% minorities, 1, 0),
    AVGPROD = if_else(HHHRWRK != 0, TOTFAMINC74 / HHHRWRK, 0)
  ) %>%
  select(
    FAMNUM, `SITE CODE`, FAMSIZE, TREAT, TOTFAMINC74, HHHRWRK, RACIALHH,
    MHETHNIC, FHETHNIC, AVGPROD, HMEVAL, VEHVAL, OTHVAL, LIQASSETS, DEBTS
  ) %>%
  rename(
```

```r
  SITE = `SITE CODE`
) %>%
mutate(SITE = case_when(
  SITE == 1 ~ "Winnipeg",
  SITE == 2 ~ "Dauphin",
  SITE == 0 ~ "Rural",
  TRUE ~ NA_character_
  )
) %>%
mutate(
  MHETHNIC = case_when(
    MHETHNIC == 1 ~ "English",
    MHETHNIC == 2 ~ "French",
    MHETHNIC == 3 ~ "German",
    MHETHNIC == 4 ~ "Irish",
    MHETHNIC == 5 ~ "Italian",
    MHETHNIC == 6 ~ "Jewish",
    MHETHNIC == 7 ~ "Native Indian band",
    MHETHNIC == 8 ~ "Native Indian non-band",
    MHETHNIC == 9 ~ "Norwegian",
    MHETHNIC == 10 ~ "Polish",
    MHETHNIC == 11 ~ "Scottish",
    MHETHNIC == 12 ~ "Ukrainian",
    MHETHNIC == 13 ~ "Metis",
    MHETHNIC == 14 ~ "Finnish",
    MHETHNIC == 15 ~ "Canadian",
    MHETHNIC == 16 ~ "Philippine",
    MHETHNIC == 17 ~ "Belgian/Dutch",
    MHETHNIC == 18 ~ "Icelandic",
    MHETHNIC == 19 ~ "Chinese",
    MHETHNIC == 20 ~ "Other",
    MHETHNIC == 21 ~ "African",
    MHETHNIC == 22 ~ "Yugoslavian/Czech/Hungarian",
    MHETHNIC == 23 ~ "Swedish",
    MHETHNIC == 24 ~ "West Indian",
    MHETHNIC == 25 ~ "Mennonite",
    MHETHNIC == 26 ~ "South American",
    MHETHNIC == 27 ~ "Latvian/Lithuanian/Estonian",
    MHETHNIC == 28 ~ "Spanish/Portuguese",
    MHETHNIC == 29 ~ "Russian",
    MHETHNIC == 30 ~ "Welsh",
    MHETHNIC == 31 ~ "Greek",
    MHETHNIC == 32 ~ "Black",
    MHETHNIC == 33 ~ "Japanese",
    MHETHNIC == 34 ~ "Danish",
    MHETHNIC == 35 ~ "American",
    TRUE ~ NA_character_
  )
) %>%
   mutate(
  FHETHNIC = case_when(
    FHETHNIC == 1 ~ "English",
    FHETHNIC == 2 ~ "French",
```

```r
      FHETHNIC == 3 ~ "German",
      FHETHNIC == 4 ~ "Irish",
      FHETHNIC == 5 ~ "Italian",
      FHETHNIC == 6 ~ "Jewish",
      FHETHNIC == 7 ~ "Native Indian band",
      FHETHNIC == 8 ~ "Native Indian non-band",
      FHETHNIC == 9 ~ "Norwegian",
      FHETHNIC == 10 ~ "Polish",
      FHETHNIC == 11 ~ "Scottish",
      FHETHNIC == 12 ~ "Ukrainian",
      FHETHNIC == 13 ~ "Metis",
      FHETHNIC == 14 ~ "Finnish",
      FHETHNIC == 15 ~ "Canadian",
      FHETHNIC == 16 ~ "Philippine",
      FHETHNIC == 17 ~ "Belgian/Dutch",
      FHETHNIC == 18 ~ "Icelandic",
      FHETHNIC == 19 ~ "Chinese",
      FHETHNIC == 20 ~ "Other",
      FHETHNIC == 21 ~ "African",
      FHETHNIC == 22 ~ "Yugoslavian/Czech/Hungarian",
      FHETHNIC == 23 ~ "Swedish",
      FHETHNIC == 24 ~ "West Indian",
      FHETHNIC == 25 ~ "Mennonite",
      FHETHNIC == 26 ~ "South American",
      FHETHNIC == 27 ~ "Latvian/Lithuanian/Estonian",
      FHETHNIC == 28 ~ "Spanish/Portuguese",
      FHETHNIC == 29 ~ "Russian",
      FHETHNIC == 30 ~ "Welsh",
      FHETHNIC == 31 ~ "Greek",
      FHETHNIC == 32 ~ "Black",
      FHETHNIC == 33 ~ "Japanese",
      FHETHNIC == 34 ~ "Danish",
      FHETHNIC == 35 ~ "American",
      TRUE ~ NA_character_
    )
  )

minc_clean$SITE <- factor(minc_clean$SITE)

colnames(minc_clean) <- toupper(colnames(minc_clean))

minc_summary <- select(minc_clean, FAMSIZE, TOTFAMINC74, HHHRWRK, RACIALHH, AVGPROD)
summary(minc_summary)
```

```
##     FAMSIZE        TOTFAMINC74       HHHRWRK        RACIALHH
##  Min.   : 1.000   Min.   :    0   Min.   :   0   Min.   :0.00000
##  1st Qu.: 2.000   1st Qu.: 4522   1st Qu.: 600   1st Qu.:0.00000
##  Median : 3.000   Median : 6825   Median : 775   Median :0.00000
##  Mean   : 3.302   Mean   : 7178   Mean   : 693   Mean   :0.05289
##  3rd Qu.: 5.000   3rd Qu.: 9203   3rd Qu.: 800   3rd Qu.:0.00000
##  Max.   :12.000   Max.   :25911   Max.   :1399   Max.   :1.00000
##  NA's   :250      NA's   :271     NA's   :588
##     AVGPROD
##  Min.   :  0.000
```

```
##  1st Qu.:  7.705
##  Median : 10.284
##  Mean   : 11.862
##  3rd Qu.: 14.294
##  Max.   :209.000
##  NA's   :598
```

1.3 Panel Data

```r
for(i in 1:11){
  payments_data[[paste0("HH_HOURS", i)]] <-
    rowSums(
      payments_data[, c(paste0("HOURS", i), paste0("FHOURS", i))],
      na.rm = TRUE
    )
}

for(i in 1:11){
  payments_data[[paste0("HH_WAGE", i)]] <-
    rowSums(
      payments_data[, c(paste0("WAGE", i), paste0("FWAGE", i))],
      na.rm = TRUE
    )
}

payments_data <- payments_data %>%
  mutate(
    GBI_LVL_YR = case_when(
      PLAN %in% c("1", "3", "6") ~ 3800,
      PLAN %in% c("2", "4", "7") ~ 4800,
      PLAN %in% c("5", "8") ~ 5800,
      PLAN == "9" ~ 0,   # control group
      TRUE ~ NA_real_
    )
    ) %>%
  mutate(
    TAX_RATE = case_when(
      PLAN %in% c("1", "2") ~ 0.35,
      PLAN %in% c("3", "4", "5") ~ 0.50,
      PLAN %in% c("6", "7", "8") ~ 0.75,
      PLAN == "9" ~ 0,
      TRUE ~ NA_real_
    )
  ) %>%
  mutate(GBI_MON = GBI_LVL_YR / 12)

payments_data <- payments_data %>%
  select(
    FAMNUM, PLAN, GBI_LVL_YR, GBI_MON, TAX_RATE, HH_HOURS1, HH_HOURS2,
    HH_HOURS3, HH_HOURS4, HH_HOURS5, HH_HOURS6, HH_HOURS7, HH_HOURS8,
    HH_HOURS9, HH_HOURS10, HH_HOURS11, HH_WAGE1, HH_WAGE2, HH_WAGE3,
    HH_WAGE4, HH_WAGE5, HH_WAGE6, HH_WAGE7, HH_WAGE8, HH_WAGE9,
    HH_WAGE10, HH_WAGE11
```

```r
  )

merged_data <- minc_clean %>%
  left_join(payments_data, by = "FAMNUM") %>%
  mutate(
    HH_WAGE0 = (mincome_data$MHTOTERN73 + mincome_data$FHTOTERN73)/12,
    HH_HOURS0 = HHHRWRK/12
  )

for(i in 1:11){
  merged_data[[paste0("PAYMENT", i)]] <-
    pmax(0,
         merged_data$GBI_MON -
           merged_data$TAX_RATE *
           merged_data[[paste0("HH_WAGE", i)]])
}
```

```r
panel_df <- merged_data %>%
  pivot_longer(
    cols = matches("HH_HOURS|HH_WAGE|PAYMENT"),
    names_to = c(".value", "period"),
    names_pattern = "(HH_HOURS|HH_WAGE|PAYMENT)(\\d+)"
  ) %>%
  mutate(period = as.numeric(period),
          TREATED = ifelse(TREAT != 9, 1, 0)) %>%
  select(
    -MHETHNIC, -FHETHNIC, -AVGPROD, -PLAN
  )
```

1.4 Exploratory Data Analysis

```r
p1 <- ggplot(minc_clean, aes(x = FAMSIZE)) +
  geom_histogram(bins = 40, na.rm = TRUE) +
  xlab("Family Size") +
  ylab("Count") +
  labs(title = "Distribution of Household Size")

p2 <- ggplot(minc_clean, aes(SITE, HHHRWRK)) +
  geom_boxplot(na.rm = TRUE) +
  xlab("Site of Experiment") +
  ylab("Household Work Hours") +
  labs(title = "Household Work Hours by Site")

p3 <- ggplot(minc_clean, aes(HHHRWRK, FAMSIZE)) +
  geom_point(alpha = 0.4, na.rm = TRUE) +
  geom_smooth(method = "lm", na.rm = TRUE) +
  xlab("Household Hours Worked") +
  ylab("Household Income") +
  labs(title = "Income vs Household Work Hours")

minc_clean %>%
  group_by(SITE) %>%
  summarise(
```

```r
    count = n(),
    avg_income = mean(TOTFAMINC74, na.rm = TRUE),
    avg_hours = mean(HHHRWRK, na.rm = TRUE),
  )
```

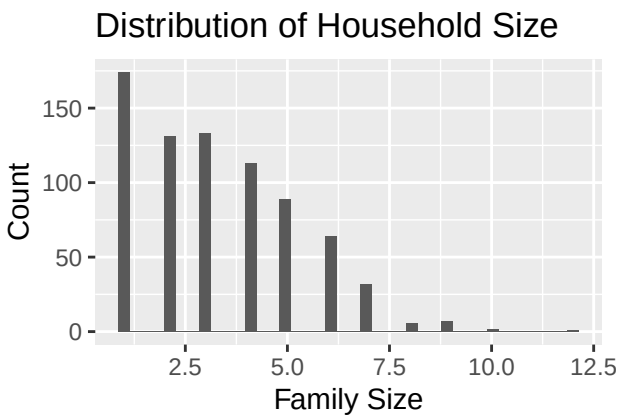
```
## # A tibble: 4 x 4
##   SITE     count avg_income avg_hours
##   <fct>    <int>      <dbl>     <dbl>
## 1 Dauphin    158      5577.      672.
## 2 Rural       75      6123.      689.
## 3 Winnipeg   519      7823.      701.
## 4 <NA>       250       NaN       NaN
```

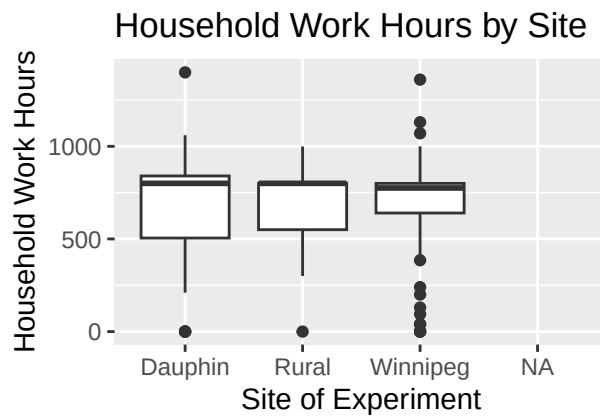
```r
breakdown <- minc_clean %>%
  count(SITE, RACIALHH)

breakdown
```

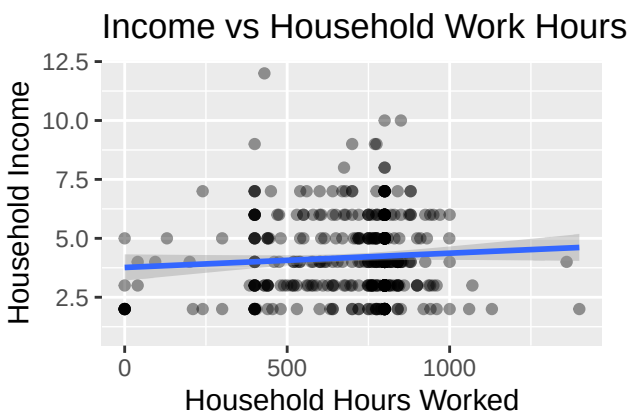
```
## # A tibble: 7 x 3
##   SITE     RACIALHH     n
##   <fct>       <dbl> <int>
## 1 Dauphin         0   156
## 2 Dauphin         1     2
## 3 Rural           0    72
## 4 Rural           1     3
## 5 Winnipeg        0   471
## 6 Winnipeg        1    48
## 7 <NA>            0   250
```

```r
plot_grid(p1, p2, p3, label_size = 12)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```







2. Model fitting

2.1 Fixed Effect Model

```
panel_df <- panel_df %>%
  mutate(post = ifelse(period > 1, 1, 0))

model_df <- panel_df %>%
  select(-RACIALHH)

fe_model <- feols(
  HH_HOURS ~ HH_WAGE + PAYMENT | FAMNUM + period,
  data = model_df,
  cluster = ~FAMNUM
)
```

```
## NOTE: 7,107 observations removed because of NA values (LHS: 6,682, RHS: 7,107).
```

```
summary(fe_model)
```

```
## OLS estimation, Dep. Var.: HH_HOURS
## Observations: 4,917
## Fixed-effects: FAMNUM: 447,  period: 11
## Standard-errors: Clustered (FAMNUM)
##          Estimate Std. Error  t value   Pr(>|t|)
## HH_WAGE  0.496139   0.106999  4.63686 4.6534e-06 ***
## PAYMENT -0.615586   0.244919 -2.51343 1.2308e-02 *
```

	(1)
HH_WAGE	0.496
	(0.107)
PAYMENT	−0.616
	(0.245)
Num.Obs.	4917
R2	0.639
R2 Adj.	0.602
R2 Within	0.128
R2 Within Adj.	0.127
AIC	70 244.3
BIC	73 228.0
RMSE	278.86
Std.Errors	by: FAMNUM
FE: FAMNUM	X
FE: period	X

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 278.9     Adj. R2: 0.602439
##                Within R2: 0.127854
```

```r
modelsummary(
  fe_model,
  statistic = "({std.error})"
)
```

2.2 Difference-in-Differences Model

```r
model_df$post <- ifelse(model_df$period > 0, 1, 0)

did_model <- feols(
  HH_HOURS ~ TREATED * post | FAMNUM + period,
  data = model_df,
  cluster = ~FAMNUM
)
```

```
## NOTES: 6,682 observations removed because of NA values (LHS: 6,682, RHS: 2,892).
##        174/0 fixed-effect singletons were removed (174 observations).
```

```
## The variables 'TREATED' and 'post' have been removed because of collinearity
## (see $collin.var).
```

```r
summary(did_model)
```

```
## OLS estimation, Dep. Var.: HH_HOURS
## Observations: 5,168
## Fixed-effects: FAMNUM: 448,  period: 12
## Standard-errors: Clustered (FAMNUM)
```

	(1)
TREATED × post	−108.525
	(47.078)
Num.Obs.	5168
R2	0.589
R2 Adj.	0.548
R2 Within	0.001
R2 Within Adj.	0.001
AIC	74 430.5
BIC	77 443.6
RMSE	296.83
Std.Errors	by: FAMNUM
FE: FAMNUM	X
FE: period	X

```
##              Estimate Std. Error  t value Pr(>|t|)
## TREATED:post -108.525    47.0784 -2.30519 0.021613 *
## ... 2 variables were removed because of collinearity (TREATED and post)
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 296.8     Adj. R2: 0.548413
##               Within R2: 7.897e-4
```

```r
modelsummary(did_model, statistic = "({std.error})")
```

2.3 Racialized vs. Nonracialized Households Model

```r
panel_df$treat_post <- panel_df$TREATED * panel_df$post
panel_df$triple_diff <- panel_df$TREATED * panel_df$post * panel_df$RACIALHH

ddd_model <- feols(
  HH_HOURS ~ treat_post + triple_diff | FAMNUM + period,
  data = panel_df,
  cluster = ~FAMNUM
)
```

```
## NOTES: 6,682 observations removed because of NA values (LHS: 6,682, RHS: 2,892).
##        174/0 fixed-effect singletons were removed (174 observations).
```

```r
summary(ddd_model)
```

```
## OLS estimation, Dep. Var.: HH_HOURS
## Observations: 5,168
## Fixed-effects: FAMNUM: 448,  period: 12
## Standard-errors: Clustered (FAMNUM)
##              Estimate Std. Error   t value Pr(>|t|)
## treat_post  -114.8216    52.3753 -2.192284 0.028874 *
## triple_diff  -33.5265    63.3346 -0.529355 0.596822
## ---
```

	Fixed Effect Model	Diff-in Diff Model	DDD Model
HH_WAGE	0.496***		
	(0.107)		
PAYMENT	−0.616*		
	(0.245)		
TREATED × post		−108.525*	
		(47.078)	
treat_post			−114.822*
			(52.375)
triple_diff			−33.526
			(63.335)
Num.Obs.	4917	5168	5168
R2	0.639	0.589	0.589
R2 Adj.	0.602	0.548	0.549
R2 Within	0.128	0.001	0.003
R2 Within Adj.	0.127	0.001	0.002
AIC	70 244.3	74 430.5	74 422.4
BIC	73 228.0	77 443.6	77 442.0
RMSE	278.86	296.83	296.53
Std.Errors	by: FAMNUM	by: FAMNUM	by: FAMNUM
FE: FAMNUM	X	X	X
FE: period	X	X	X

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 296.5     Adj. R2: 0.549205
##                Within R2: 0.002753
```

```r
modelsummary(
  list(
    "Fixed Effect Model" = fe_model,
    "Diff-in Diff Model" = did_model,
    "DDD Model" = ddd_model
  ),
  stars = TRUE
)
```

3. Robustness Checks

3.1 Robust SE

```r
fe_model_robust <- feols(
  HH_HOURS ~ HH_WAGE + PAYMENT | FAMNUM + period,
  data = panel_df,
  vcov = "hetero"
```

```r
)
```

```
## NOTE: 7,107 observations removed because of NA values (LHS: 6,682, RHS: 7,107).
```

```r
did_model_robust <- feols(
  HH_HOURS ~ TREATED:post | FAMNUM + period,
  data = panel_df,
  vcov = "hetero"
)
```

```
## NOTES: 6,682 observations removed because of NA values (LHS: 6,682, RHS: 2,892).
##        174/0 fixed-effect singletons were removed (174 observations).
```

```r
ddd_model_robust <- feols(
  HH_HOURS ~ TREATED:post + TREATED:post:RACIALHH | FAMNUM + period,
  data = panel_df,
  vcov = "hetero"
)
```

```
## NOTES: 6,682 observations removed because of NA values (LHS: 6,682, RHS: 2,892).
##        174/0 fixed-effect singletons were removed (174 observations).
```

```r
modelsummary(
  list(
    "FE Robust" = fe_model_robust,
    "DiD Robust" = did_model_robust,
    "DDD Robust" = ddd_model_robust
  ),
  stars = TRUE
)
```

Appendix

Save Files

```r
write.csv(minc_clean, "../data/02-clean/mincome_clean.csv", row.names = TRUE)
write.csv(merged_data, "../data/02-clean/merged_panel_data.csv", row.names = TRUE)
write.csv(panel_df, "../data/02-clean/panel_df.csv", row.names = TRUE)
```

```r
event_model <- feols(
  HH_HOURS ~ i(period, TREATED, ref = 0) | FAMNUM + period,
  data = panel_df,
  cluster = ~FAMNUM
)
```

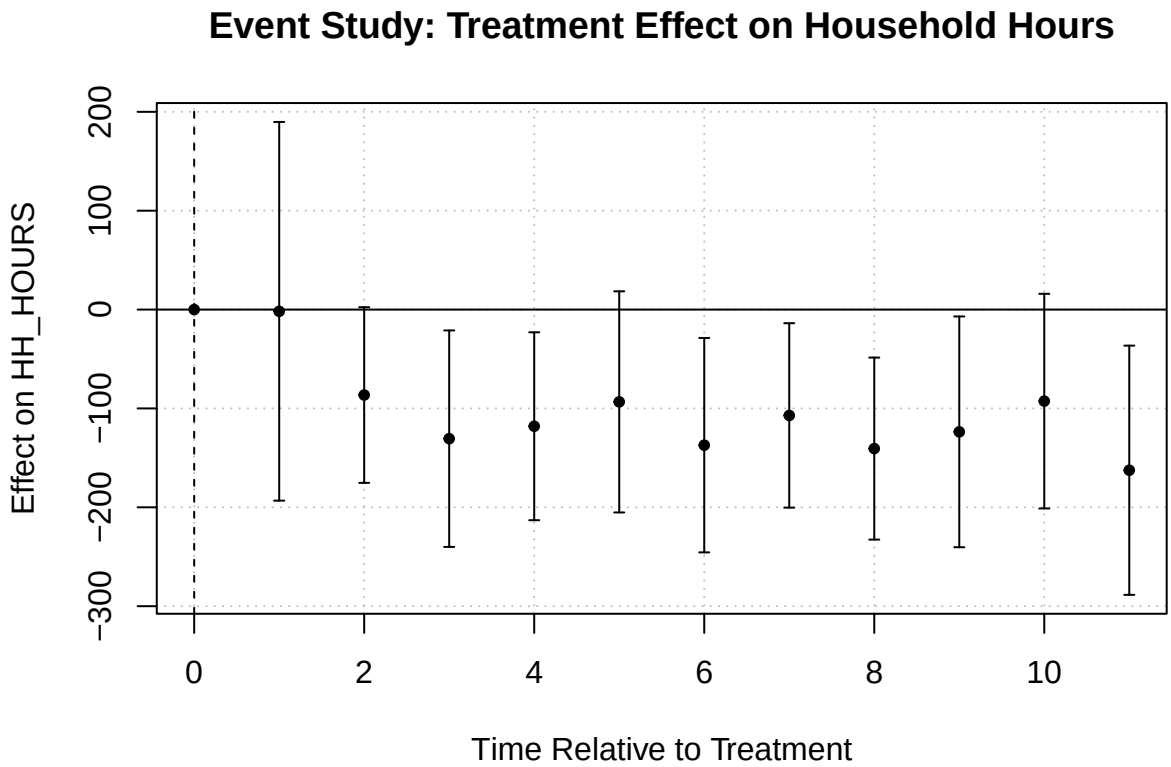
```
## NOTES: 6,682 observations removed because of NA values (LHS: 6,682, RHS: 2,892).
##        174/0 fixed-effect singletons were removed (174 observations).
```

```r
summary(event_model)
```

```
## OLS estimation, Dep. Var.: HH_HOURS
## Observations: 5,168
## Fixed-effects: FAMNUM: 448,  period: 12
## Standard-errors: Clustered (FAMNUM)
##                       Estimate Std. Error   t value  Pr(>|t|)
## period::1:TREATED     -1.75143    97.4481 -0.017973 0.9856685
## period::2:TREATED    -86.43088    45.2003 -1.912177 0.0564924 .
```

	FE Robust	DiD Robust	DDD Robust
HH_WAGE	0.496***		
	(0.094)		
PAYMENT	−0.616**		
	(0.205)		
TREATED × post		−118.011*	−114.822*
		(53.091)	(53.570)
TREATED × post × RACIALHH			−33.526
			(69.082)
Num.Obs.	4917	5168	5168
R2	0.639	0.589	0.589
R2 Adj.	0.602	0.549	0.549
R2 Within	0.128	0.003	0.003
R2 Within Adj.	0.127	0.002	0.002
AIC	70 244.3	74 420.9	74 422.4
BIC	73 228.0	77 434.0	77 442.0
RMSE	278.86	296.55	296.53
Std.Errors	Heteroskedasticity-robust	Heteroskedasticity-robust	Heteroskedasticity-robust
FE: FAMNUM	X	X	X
FE: period	X	X	X

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

```
## period::3:TREATED  -130.58054    55.6972 -2.344471 0.0194909 *
## period::4:TREATED  -118.01599    48.3540 -2.440666 0.0150475 *
## period::5:TREATED   -93.36050    56.9107 -1.640473 0.1016104
## period::6:TREATED  -137.15060    55.1692 -2.486000 0.0132831 *
## period::7:TREATED  -107.07293    47.4523 -2.256434 0.0245258 *
## period::8:TREATED  -140.61748    46.8417 -3.001973 0.0028327 **
## period::9:TREATED  -123.66504    59.3864 -2.082380 0.0378759 *
## period::10:TREATED  -92.62243    55.2338 -1.676917 0.0942578 .
## period::11:TREATED -162.50261    64.1316 -2.533894 0.0116208 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 296.4     Adj. R2: 0.548658
##               Within R2: 0.003452
```

```r
iplot(event_model,
      main = "Event Study: Treatment Effect on Household Hours",
      xlab = "Time Relative to Treatment",
      ylab = "Effect on HH_HOURS")
```